

Servlet vs JSP

👉 JavaServer Pages (JSP):

- **Purpose:** Primarily for creating dynamic web content by embedding Java code into HTML pages.
- **Syntax:** Combines HTML with Java code, using special tags (e.g., `<% %>` for scriptlets, `<%= %>` for expressions).
- **Ease of Use:** More user-friendly for designing web pages as it separates the presentation logic from business logic.
- **Execution:** JSPs are compiled into servlets by the server at runtime, which then handles the request.
- **Maintenance:** Easier to maintain and update the look and feel as the logic and presentation are typically separated.
- **Use Case:** Ideal for view components and pages with mostly HTML content, where Java code is used minimally.

👉 Servlet:

- **Purpose:** Used for handling HTTP requests and generating dynamic responses. They provide a programmatic way to create web applications.
- **Syntax:** Written in pure Java, handling both request processing and response generation within a single class.
- **Ease of Use:** More complex as it requires manual handling of HTTP requests, responses, and HTML generation.
- **Execution:** Servlets are Java classes that run on the server and are responsible for generating responses, usually returning HTML content.

- **Performance:** Servlets can be more efficient for scenarios requiring extensive server-side processing, as they avoid the overhead of JSP compilation.
- **Maintenance:** Can be more difficult to maintain due to intertwined request handling and HTML generation logic.
- **Use Case:** Suitable for complex business logic and handling tasks like database interactions, session management, and processing form data.